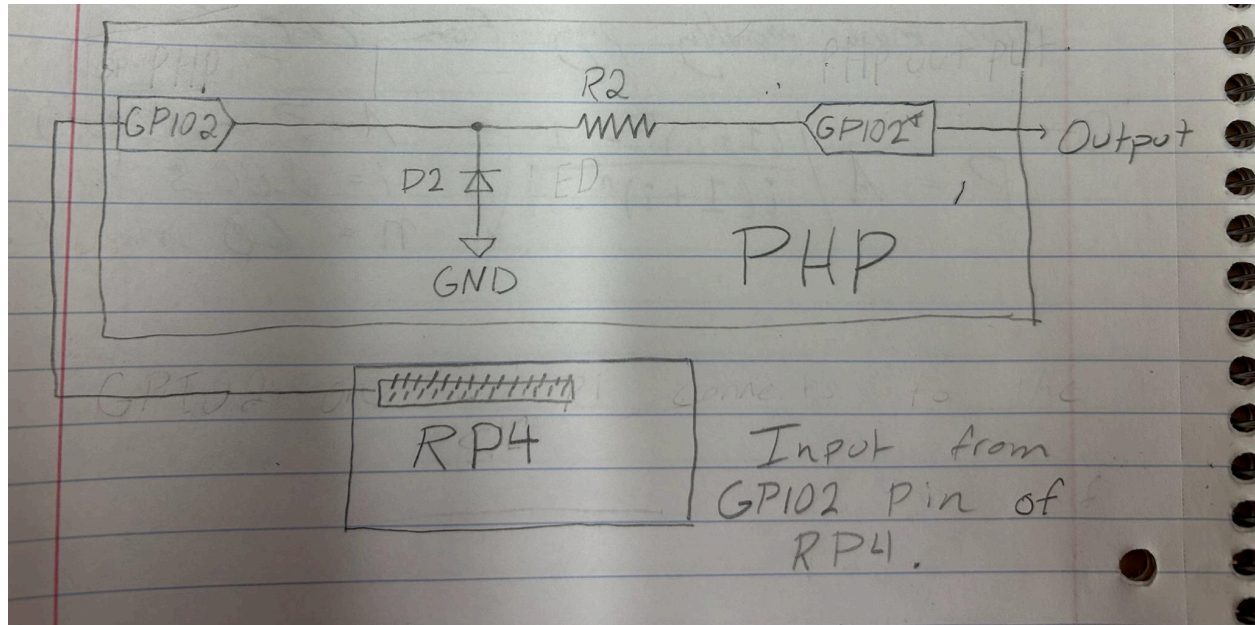




To avoid shorting the RP4, we connect the PHP to the RP4 pins.

The PHP adds a resistor in series and a parallel circuit including a diode between ground and GPIO2.

The resistor protects GPIO2 by limiting the current flowing through GPIO2.

The diode/GND parallel circuit protects GPIO2 by shunting damaging voltages to ground. Because of the direction of the diode, any negative voltages will connect to ground and if a positive voltage much higher than 3.3V is connected, the diode will short, shunting the current to ground.